Iterative Prompting

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Iterative Prompting before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Examples and Rubrics

Without opening the old lesson, explain one key idea from Examples and **Rubric**s. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Sampling and Variability

Without opening the old lesson, explain one key idea from Sampling and Variability. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Predictive and Generative AI

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Iterative Prompting trace and record the first step you can explain without help.

Use W2: Trace the worked case for Controlled prompt experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Iterative Prompting explanation against the supplied facts before discussing it.

Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

A factual response uses four paragraphs when the contract requires three bullets. Design a controlled prompt revision and evaluation.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Iterative Prompting target to a trace and a justified conclusion.

A summary has accurate facts but uses unexplained terms. Propose one prompt change and a comparison that checks both accessibility and factual preservation.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Iterative Prompting case and show the deciding evidence.

A factually correct explanation is too difficult for a nine-year-old.

Make and test one focused revision.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

You change a prompt's audience, length and source rules together; the result improves. What can you conclude, and how would you investigate the cause?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 7 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Iterative Prompting adds to Verified Study Buddy and keep the evidence.

Maintain prompt **version**, diagnosed failure, change, and before/after **rubric** scores.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A summary has accurate facts but uses unexplained terms. Propose one prompt change and a comparison that checks both accessibility and factual preservation.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Iterative Prompting facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Iterative Prompting facts and keep your reasoning.

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit the limit

> Audit the limit. Apply the supplied Iterative Prompting facts and keep your reasoning.

A report omits this qualification: Prompt selection on development cases cannot be reported as independent final performance. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Iterative Prompting facts and keep your reasoning.

On eight **paired case**s A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Iterative Prompting facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Iterative Prompting

> A fictional Study Buddy prompt invents a definition when approved notes omit it. Explain the labelled step before checking the changed condition.

A fictional Study Buddy prompt invents a definition when approved notes omit it.

P asks “answer helpfully”. Revision Q adds “if approved notes lack the answer, say not supplied”.

Hold model settings and four development questions fixed. P passes 1,2; Q passes 1,2,3. Case 4 still fails.

AI ACADEMY / AI-3/7

Iterative Prompting

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Locate failure

The target defect is unsupported answers when notes are missing.



2. One controlled change

Q adds the missing-evidence rule; other instructions remain fixed.



3. Paired results

P=2/4; Q=3/4; gain is case 3, with case 4 unresolved.



4. Next action

Inspect case 4 and reserve genuinely fresh questions for final evaluation.

CHANGE ONE THING

Change model settings and note wording at the same time as the prompt.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What **evidence** supports improvement here, and what remains unproven?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Maintain a Study Buddy experiment log linking failure, single revision and paired **rubric** results.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Make a transparent first **version**. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Maintain a Study Buddy experiment log linking failure, single revision and paired **rubric** results.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy prompt invents a definition when approved notes omit it.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Inspect case 4 and reserve genuinely fresh questions for final evaluation.

1. Locate failure

2. Next action

3. One controlled change

4. Paired results

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Next action Inspect case 4 and reserve genuinely fresh questions for final evaluation. An **iteration** should respond to a recorded failure with a precise change.

Comparing the same questions reveals whether other behavior changed too. The invented results show a development gain of one case, not a guarantee that the new instruction always works.

Save prompt text, settings, source notes and case-level outcomes.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change model settings and note wording at the same time as the prompt. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The comparison no longer isolates the prompt revision.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The comparison no longer isolates the prompt revision. Several changes could explain the altered **output**s; record them or run separate controlled comparisons.

Repair a missing building block

Review Examples and **Rubric**s (ai-3/6). Goal: Write a prompt specification with observable response criteria. Short practice: Explain why neither original nor changed draft is a full pass, and provide a complete repaired card..

Return to this week and explain what you changed.

Review Sampling and Variability (ai-3/4). Goal: Sample from a distribution using explicit cumulative intervals. Short practice: Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?.

Return to this week and explain what you changed.

Review Predictive and Generative AI (ai-3/1). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Identify the first unsupported claim and explain what source would be needed before asserting it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

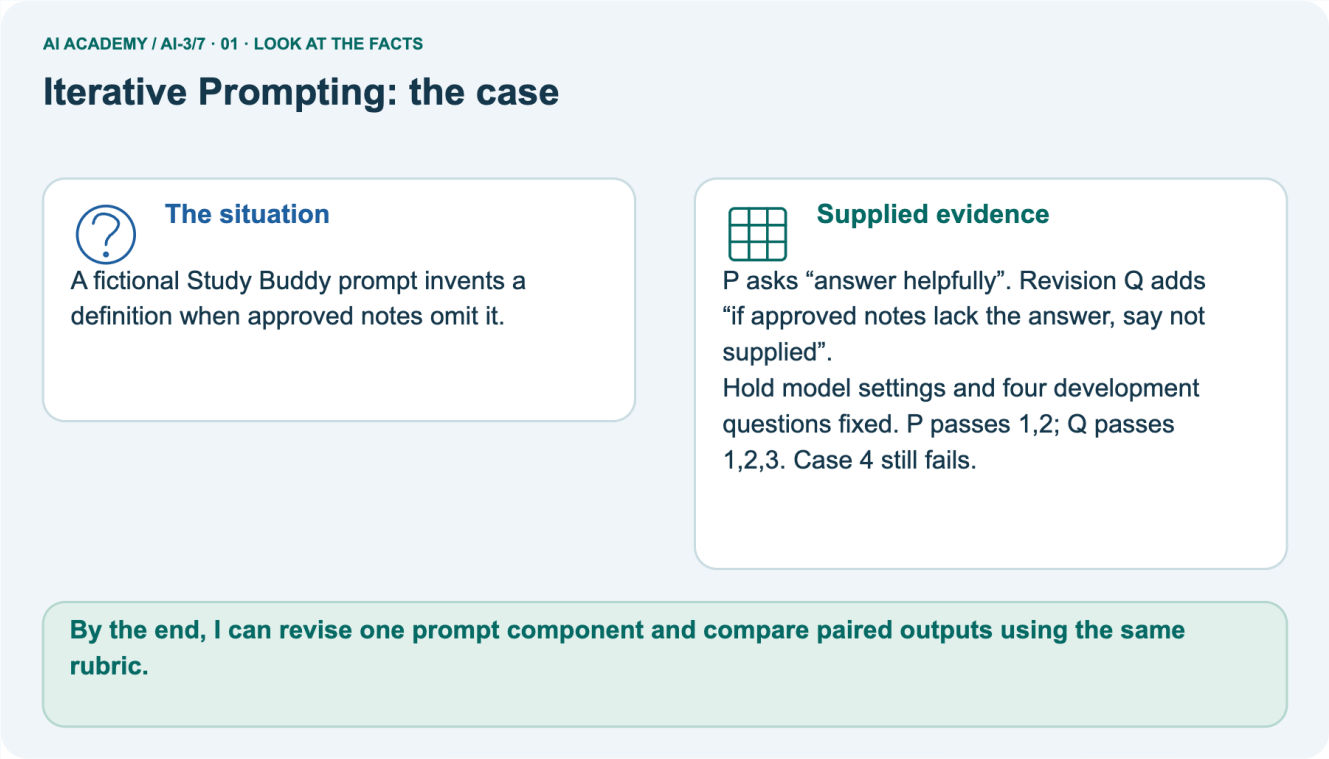
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Iterative Prompting picture from its labels, then record a relevant limit.

01. Iterative Prompting: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

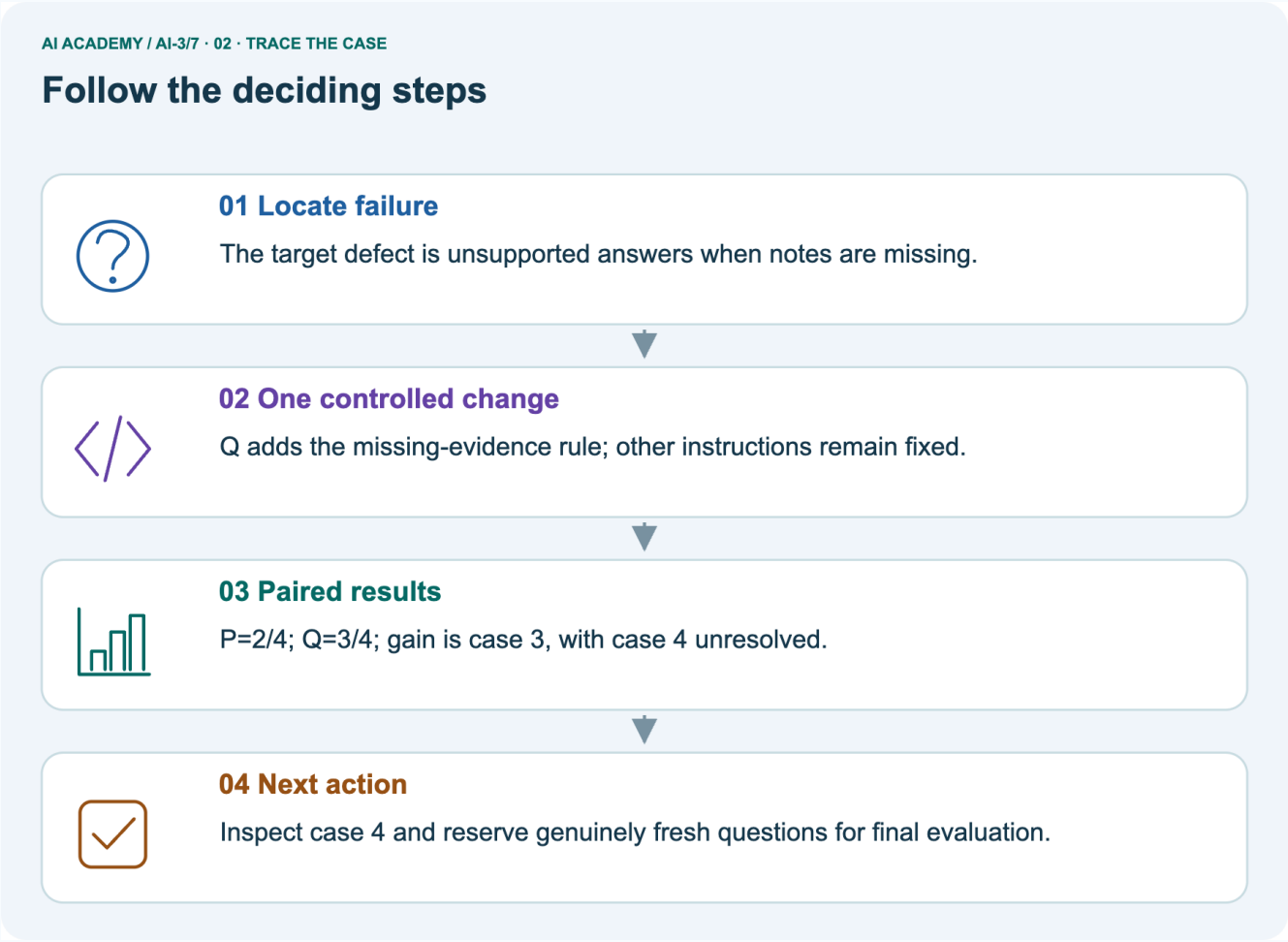
A fictional Study Buddy prompt invents a definition when approved notes omit it. Supplied facts: P asks “answer helpfully”. Revision Q adds “if approved notes lack the answer, say not supplied”.

Hold model settings and four development questions fixed. P passes 1,2; Q passes 1,2,3. Case 4 still fails.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



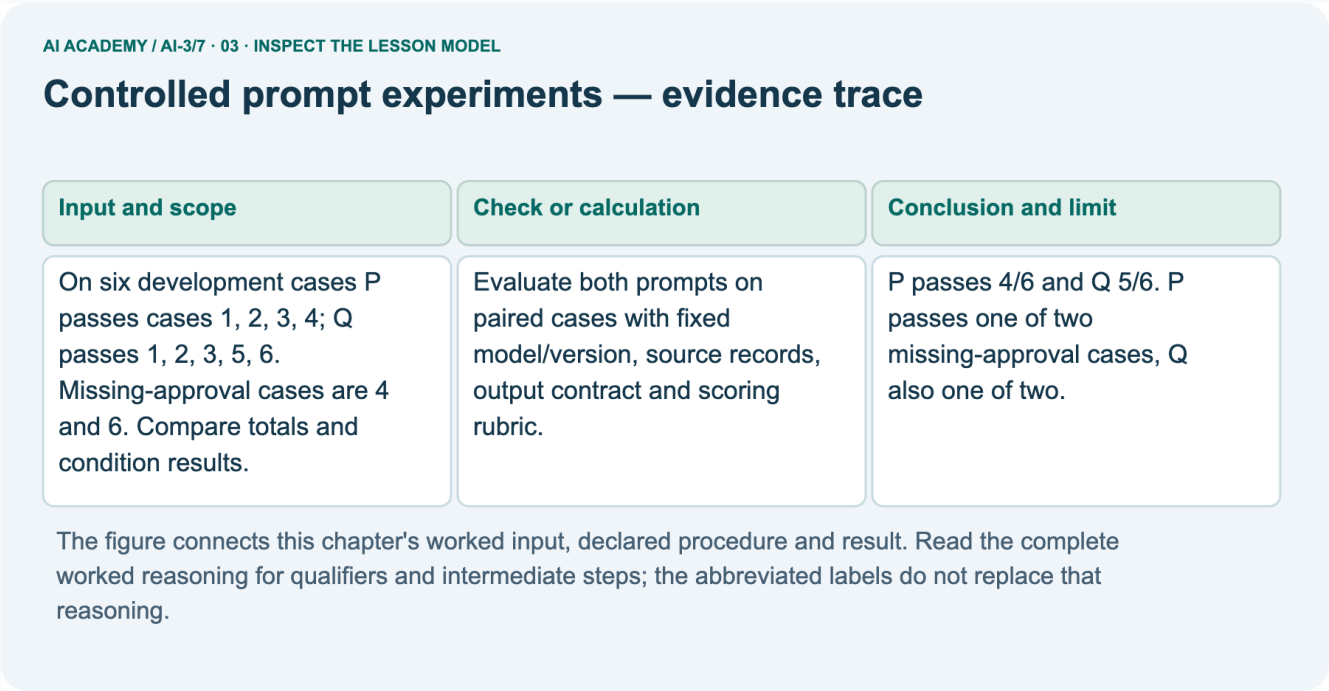
Follow the labelled picture with the matching explanation. Open the vector diagram

Locate failure: The target defect is unsupported answers when notes are missing. → One controlled change: Q adds the missing-**evidence** rule; other instructions remain fixed. → Paired results: P=2/4; Q=3/4; gain is case 3, with case 4 unresolved. → Next action: Inspect case 4 and reserve genuinely fresh questions for final evaluation.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Controlled prompt experiments — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Controlled prompt experiments — **evidence** trace. On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6.

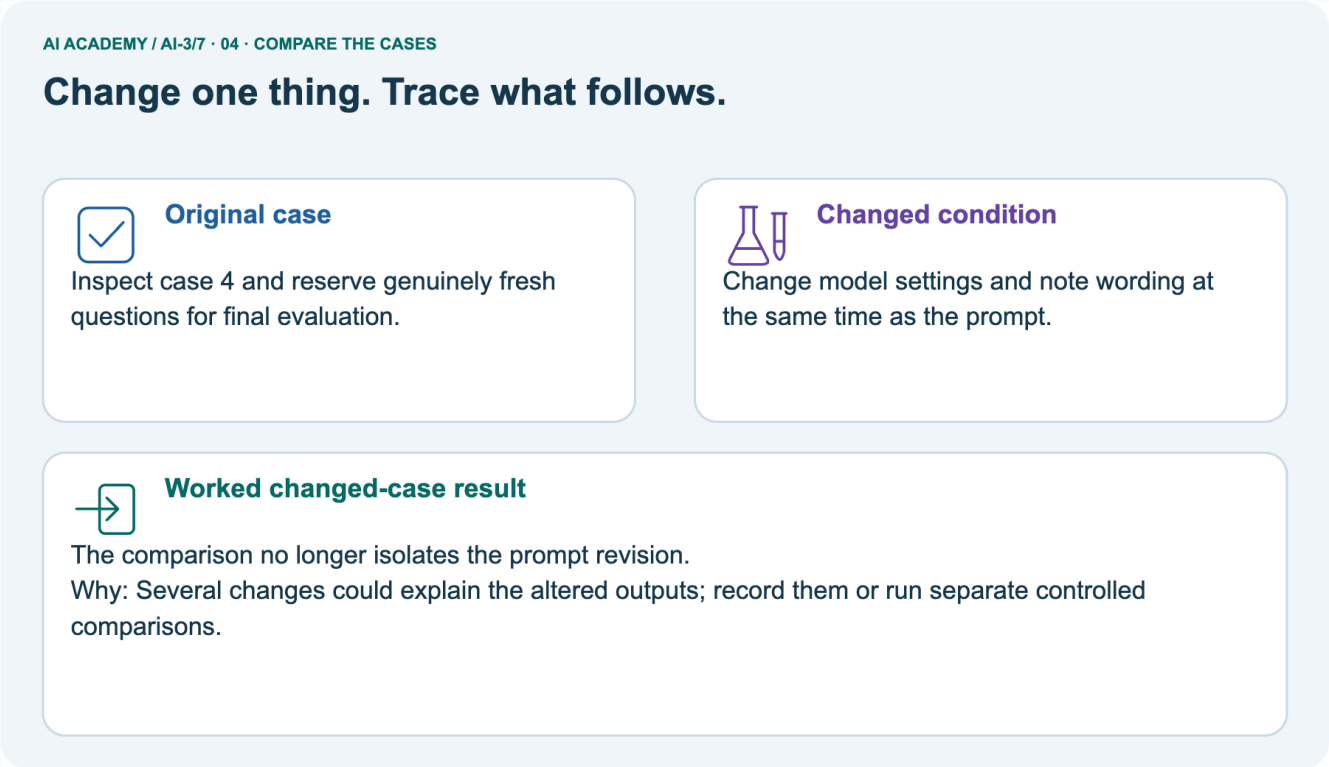
Compare totals and condition results. | Evaluate both prompts on **paired case**s with fixed model/**version**, source records, **output** contract and scoring **rubric**. | P passes 4/6 and Q 5/6.

P passes one of two missing-approval cases, Q also one of two.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

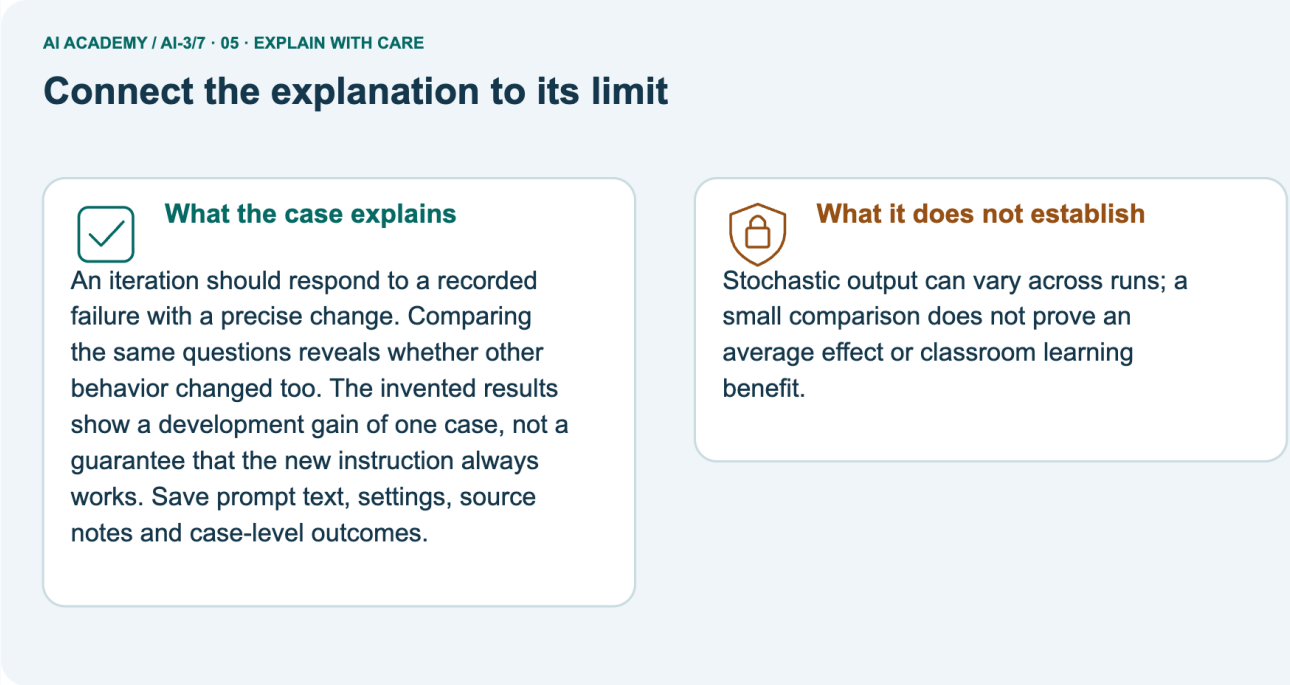
Original case: Inspect case 4 and reserve genuinely fresh questions for final evaluation. Changed condition: Change model settings and note wording at the same time as the prompt.

Expected result: The comparison no longer isolates the prompt revision. Reason: Several changes could explain the altered **output**s; record them or run separate controlled comparisons.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An **iteration** should respond to a recorded failure with a precise change. Comparing the same questions reveals whether other behavior changed too.

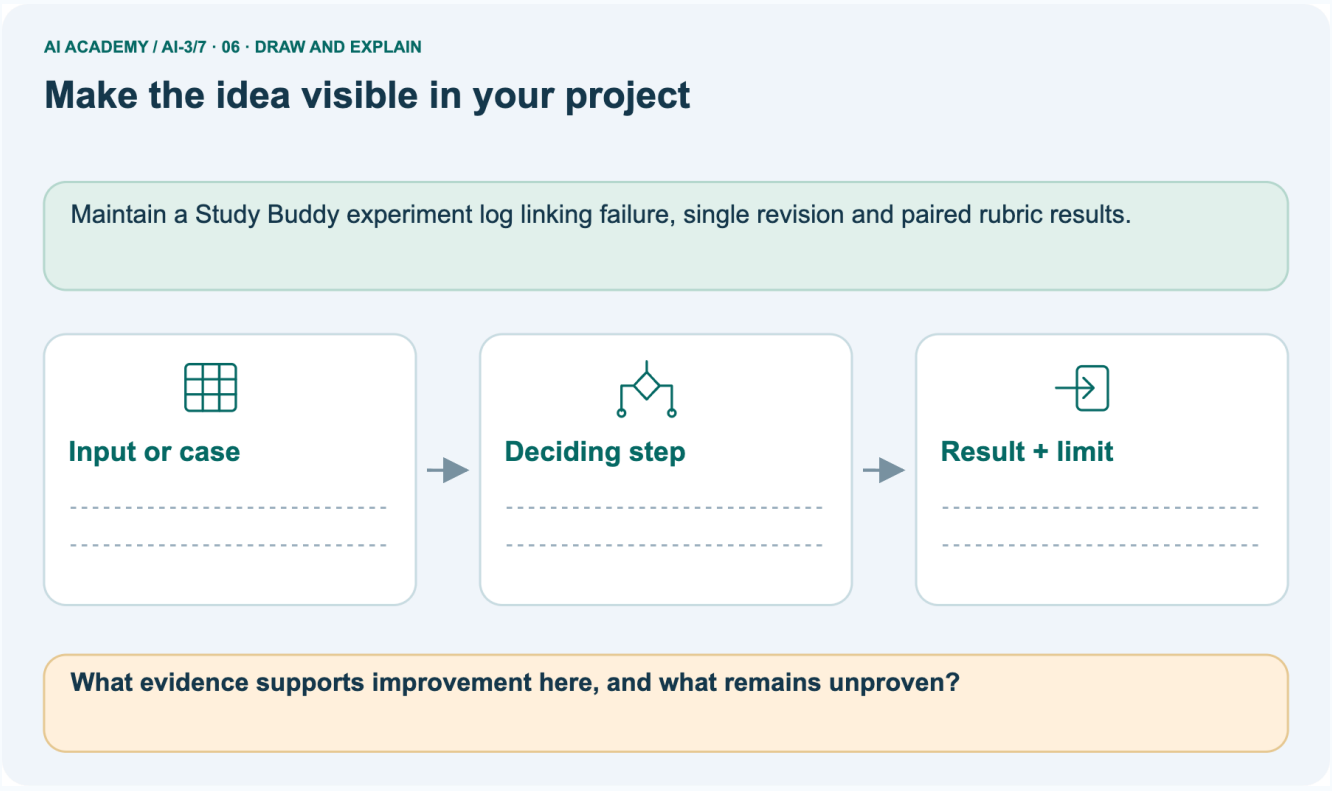
The invented results show a development gain of one case, not a guarantee that the new instruction always works. Save prompt text, settings, source notes and case-level outcomes.

Limit: Stochastic **output** can vary across runs; a small comparison does not prove an average effect or classroom learning benefit.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain a Study Buddy experiment log linking failure, single revision and paired **rubric** results. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** supports improvement here, and what remains unproven?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.